



Republic of the Philippines
BATANGAS STATE UNIVERSITY

The National Engineering University

Lipa Campus

A. Tanco Drive, Brgy. Marawoy, Lipa, Batangas, Philippines 4217

Tel Nos.: (+63 43) 980-0385; 980-0387; 980-0392 to 94 loc. 3130

E-mail Address: cics.lipa@g.batstate-u.edu.ph | Website Address: <http://www.batstate-u.edu.ph>

College of Informatics and Computing Sciences

ASIA Semestral Project – Material 3: Technical Integration Registry

Objective: To document the technical "interoperability" of your prototype and verify that your system architecture supports professional integration standards.

Step 1: The Integration Inventory

List **ALL** individual integration points. Classify them using the standardized categories provided.

Standard Categories:

- **Web API:** Services accessed via HTTP (e.g., Google OAuth, Payment Gateways, SMS APIs).
- **Hardware:** Physical devices sending data to your code (e.g., RFID, Biometric Scanners, Sensors).
- **External Data:** Importing files created outside your system (e.g., Legacy JSON/CSV migrations).
- **Middleware:** Software layers facilitating communication (e.g., RabbitMQ, custom Bridge scripts).
- **Cloud Service:** Direct cloud-native integrations (e.g., AWS S3, Firebase Firestore).

Integration Name	Category	Specific Purpose	Technical Implementation Details
Example: Semaphore	Web API	SMS Notifications	The Business Tier triggers a POST request to <code>api.semaphore.co</code> containing the student's mobile number.
Example: RC522 RFID	Hardware	Attendance Entry	The physical reader scans the UID. A local script captures the serial data and sends it to the Backend Controller.

Step 2: Technical Layer Mapping

Describe the architectural layers of your project. This must match the folder structure shown in your later evidence screenshots.

- **Project Title:** [Insert Project Name]
- **Layer 1 (The Interface):** (e.g., "React.js Web Dashboard").
- **Layer 2 (The Logic):** (e.g., "Node.js scripts in the `/server` directory").
- **Layer 3 (The Storage):** (e.g., "MySQL via XAMPP").
- **Integrated Data Flow:** "When a user performs [Action] in Layer 1, it sends a request to Layer 2, which then communicates with [Integration Name from Step 1] and updates the records in Layer 3."



Republic of the Philippines
BATANGAS STATE UNIVERSITY

The National Engineering University

Lipa Campus

A. Tanco Drive, Brgy. Marawoy, Lipa, Batangas, Philippines 4217

Tel Nos.: (+63 43) 980-0385; 980-0387; 980-0392 to 94 loc. 3130

E-mail Address: cics.lipa@g.batstate-u.edu.ph | Website Address: <http://www.batstate-u.edu.ph>

College of Informatics and Computing Sciences

Step 3: Mandatory Technical Evidence (Procedures)

You must backup every claim with a screenshot. Follow these specific steps to capture your evidence.

Evidence 1: Integration Verification

A. If using Web APIs (Postman Procedure):

1. **Open Postman** and create a new request tab.
2. **Enter the URL:** Paste your specific API endpoint (e.g., `https://api.semaphore.co/v4/messages`).
3. **Set the Method:** Select `GET` or `POST` as required by your integration.
4. **Authorize:** In the "Auth" or "Headers" tab, enter your API Key.
5. **Click Send:** Ensure you receive a response.
6. **Capture:** The screenshot must clearly show the **green "Status: 200 OK"** and the **JSON response** in the bottom window.

B. If using Hardware (Serial Monitor Procedure):

1. **Connect Device:** Plug your hardware (Arduino/ESP32) into your computer.
2. **Open IDE:** Open the Arduino IDE and go to `Tools > Serial Monitor`.
3. **Trigger Action:** Perform a physical action (e.g., swipe an RFID card or heat a sensor).
4. **Capture:** The screenshot must show the **live log** printing the data value (e.g., `UID: 5A33 B2`) and a success confirmation.

C. If using External Data (Console Procedure):

1. **Run Script:** Execute your import/migration script in your IDE (VS Code/Eclipse).
2. **Locate Console:** Open the **Terminal or Debug Console** at the bottom of the editor.
3. **Capture:** The screenshot must show the specific log line confirming data movement (e.g., `Successfully parsed 25 records from external_data.json`).

Evidence 2: The Architectural Structure (Folder Proof)

- **Action:** Open your code editor and expand your file browser sidebar.
- **Capture:** The sidebar clearly showing separate folders for your UI/Frontend, Logic/Backend, and Data/Configurations. Highlight the file that handles the integration mentioned in Step 1.

Evidence 3: Data Integrity & Security Check

- **Action:** Open your Database Manager (e.g., MySQL Workbench) and run `SELECT * FROM users`.
- **Capture:** The results window showing the **Password** column.
- **Verification:** The passwords must appear as **Hashed/Encrypted strings** (e.g., `$2y$10...`) and not plain text.
- **Note:** If no login exists, capture your `.env` or config file showing that API Keys are stored in a separate, secure file.



Republic of the Philippines
BATANGAS STATE UNIVERSITY

The National Engineering University

Lipa Campus

A. Tanco Drive, Brgy. Marawoy, Lipa, Batangas, Philippines 4217

Tel Nos.: (+63 43) 980-0385; 980-0387; 980-0392 to 94 loc. 3130

E-mail Address: cics.lipa@g.batstate-u.edu.ph | Website Address: <http://www.batstate-u.edu.ph>

College of Informatics and Computing Sciences

Meeting Deliverable:

By the end of the session, students must have completed the **Inventory Table**, the **Layer Mapping**, and provided the **3 Labeled Screenshots** as proof of implementation.